

Evaluation Statistics

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1 Setup

1.1 Import and Filter Data

Table 1: Entries excluded by timeout and dependency filtering.

phase	removal_reason	removed_entries
Phase 5	Elapsed time exceeded configured timeout plus grace period	5
R1	Elapsed time exceeded configured timeout plus grace period	10
R3	Elapsed time exceeded configured timeout plus grace period	3
R3	Recovery execution below 10 seconds; treated as not run	2
R4	Elapsed time exceeded configured timeout plus grace period	3
R5	Dependent recovery or stability entry removed with primary phase	3
R5	Elapsed time exceeded configured timeout plus grace period	2
R5	Recovery execution below 10 seconds; treated as not run	89
SR	Dependent recovery or stability entry removed with primary phase	3

1.2 Theme Settings

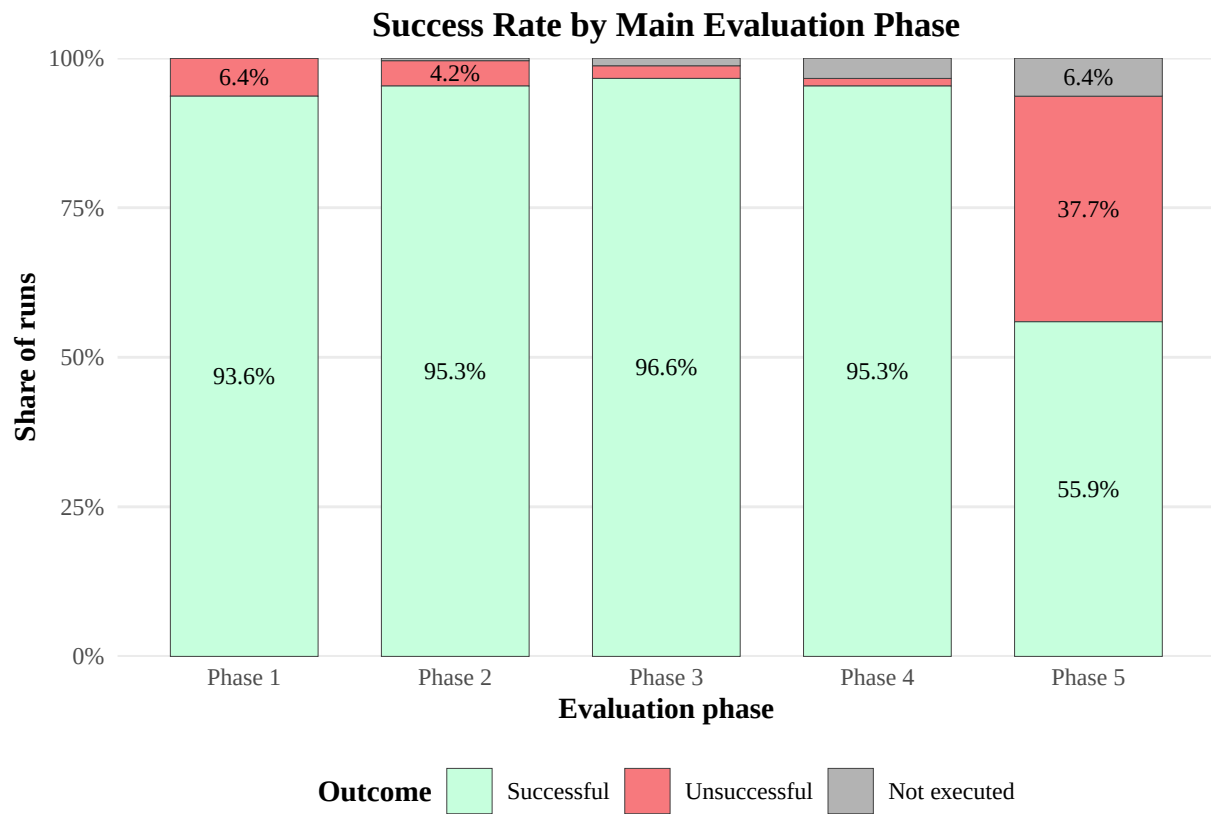
2 Success Rates

Table 2: Number of Valid Runs Recorded for Each Evaluation Phase

Phase	Number of runs
Phase 1	236
S1	106
R1	120
Phase 2.1	235
Phase 2.2	234
Phase 3	233
S3	222

Phase	Number of runs
R3	6
Phase 4	228
S4	133
R4	92
Phase 5	221
S5	107
R5	25
SR	238

2.1 Final Outcome by Main Phase



2.2 Direct, Recovered, Failed, and Not-Run Outcomes

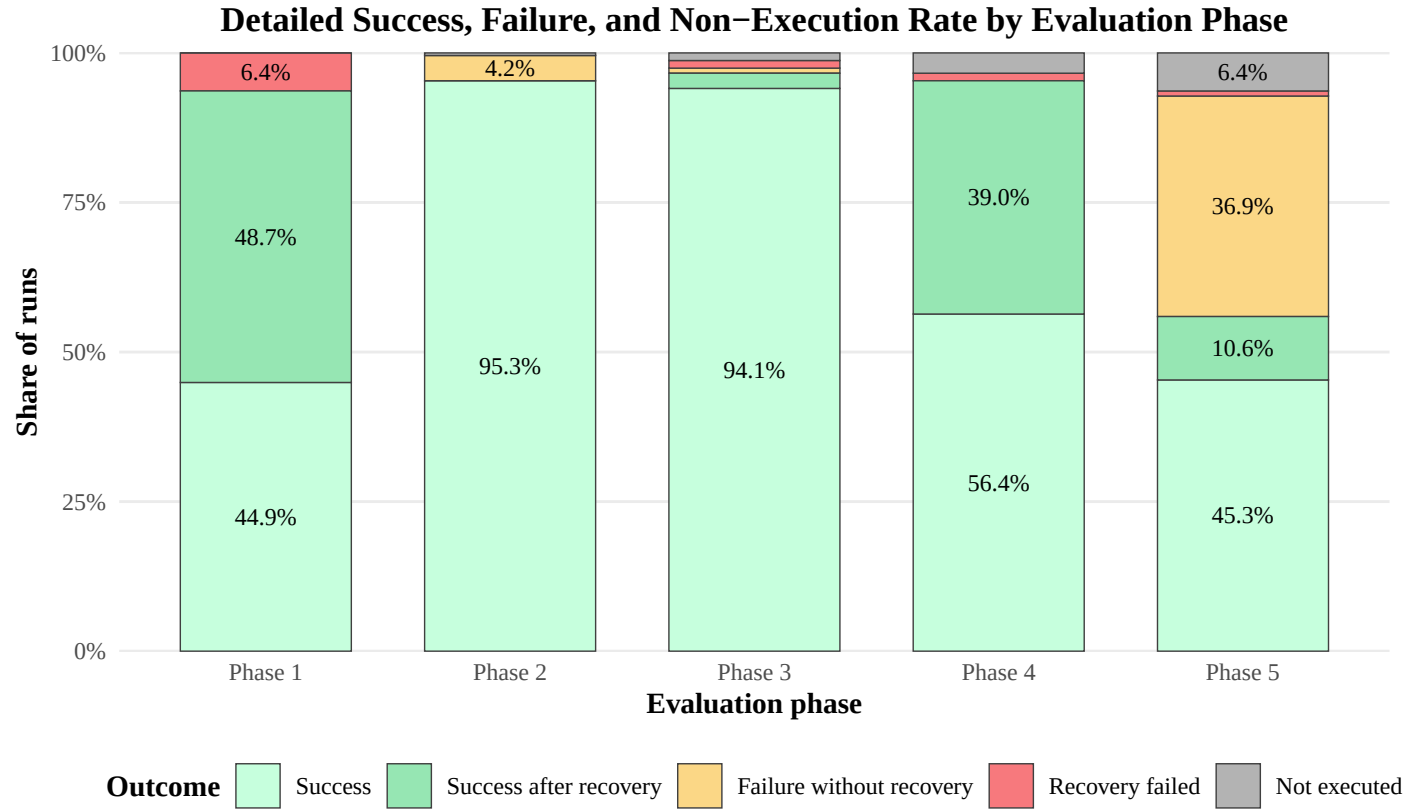


Table 3: Run counts by phase outcome.

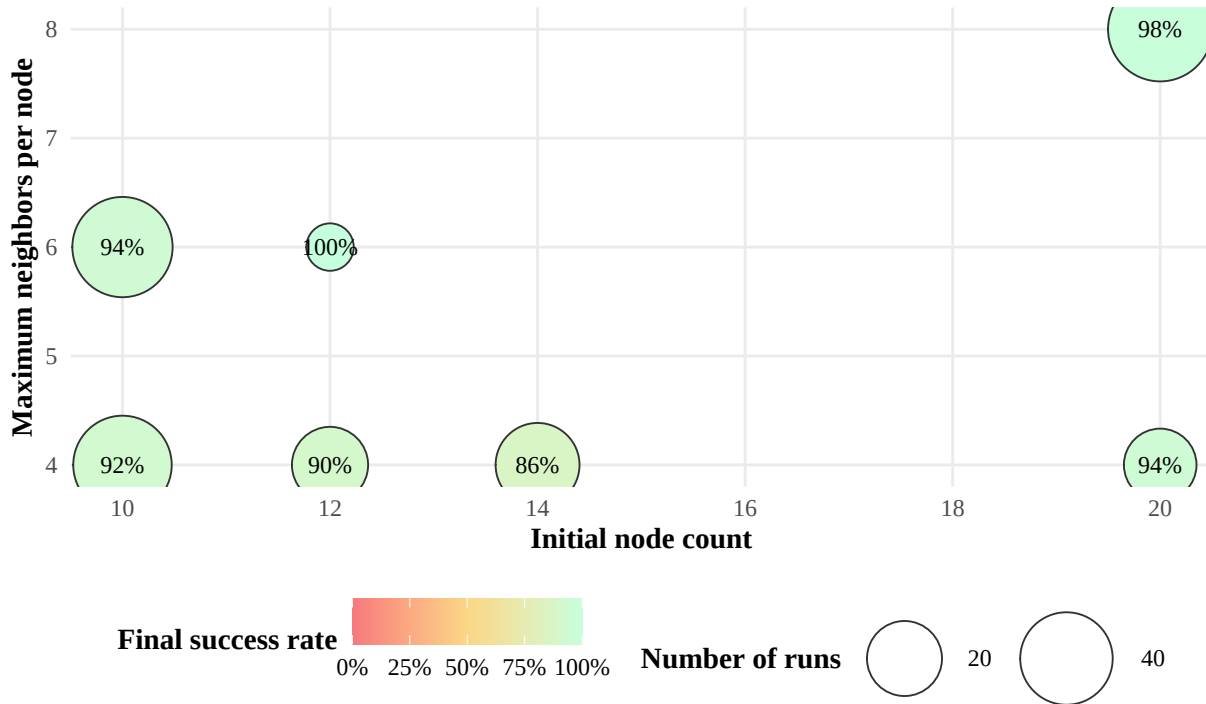
phase_group	Success (no recovery)	Success (with recovery)	Failure (no recovery)	Failure (with recovery)
Phase 1	106	115	0	15
Phase 2	225	0	10	0
Phase 3	222	6	2	3
Phase 4	133	92	0	3
Phase 5	107	25	87	2

3 Final Success Rate by Input Variables

3.1 Phase 1: Initial Network Formation

Phase 1 Initial-Network Formation: Final Success Rate by Network Size and Neighbor

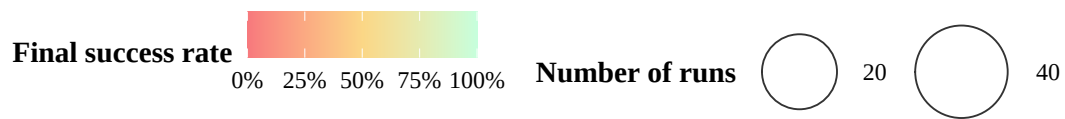
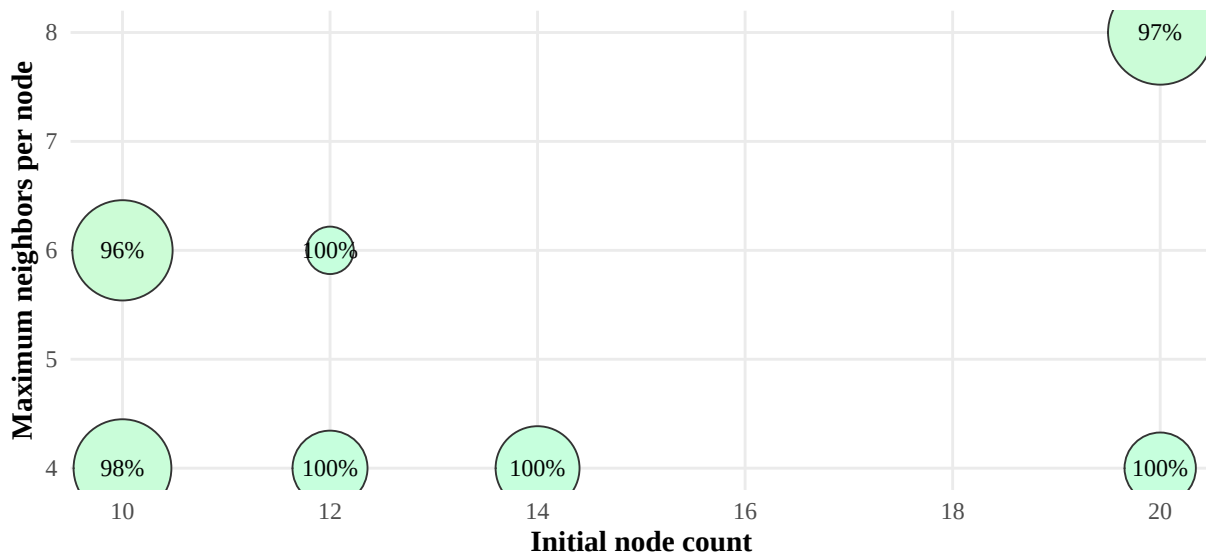
Bubble fill and label show final success rate including successful R1 recovery; bubble size shows the number of ru



3.2 Phase 3: Single-Node Failure Repair

ase 3 Single-Node Failure Repair: Final Success Rate by Network Size and Neighbor

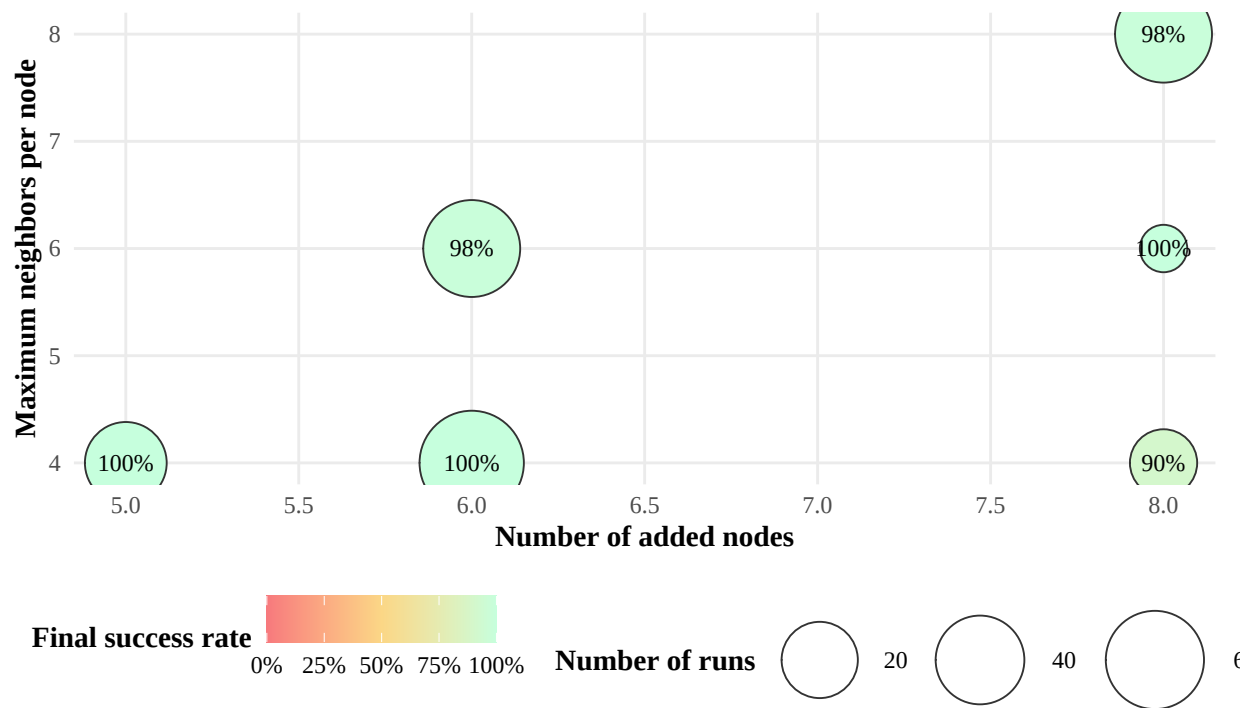
Bubble fill and label show final success rate including successful R3 recovery; bubble size shows the number of ru



3.3 Phase 4: Adding Nodes

Phase 4 Node Addition: Final Success Rate by Added Nodes and Neighbor Limit

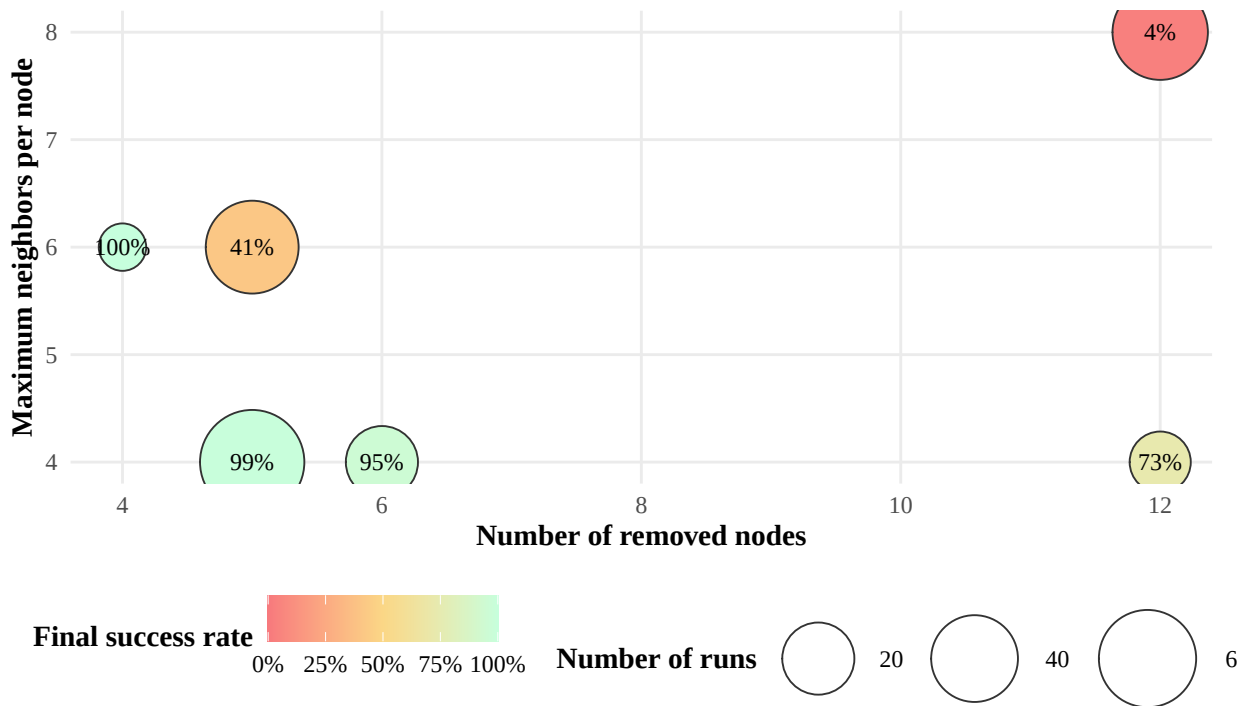
Bubble fill and label show final success rate including successful R4 recovery; bubble size shows the number of ru



3.4 Phase 5: Removing Multiple Nodes

Figure 5 Multiple-Node Failure Repair: Final Success Rate by Removed Nodes and Neigh

Bubble fill and label show final success rate including successful R5 recovery; bubble size shows the number of ru



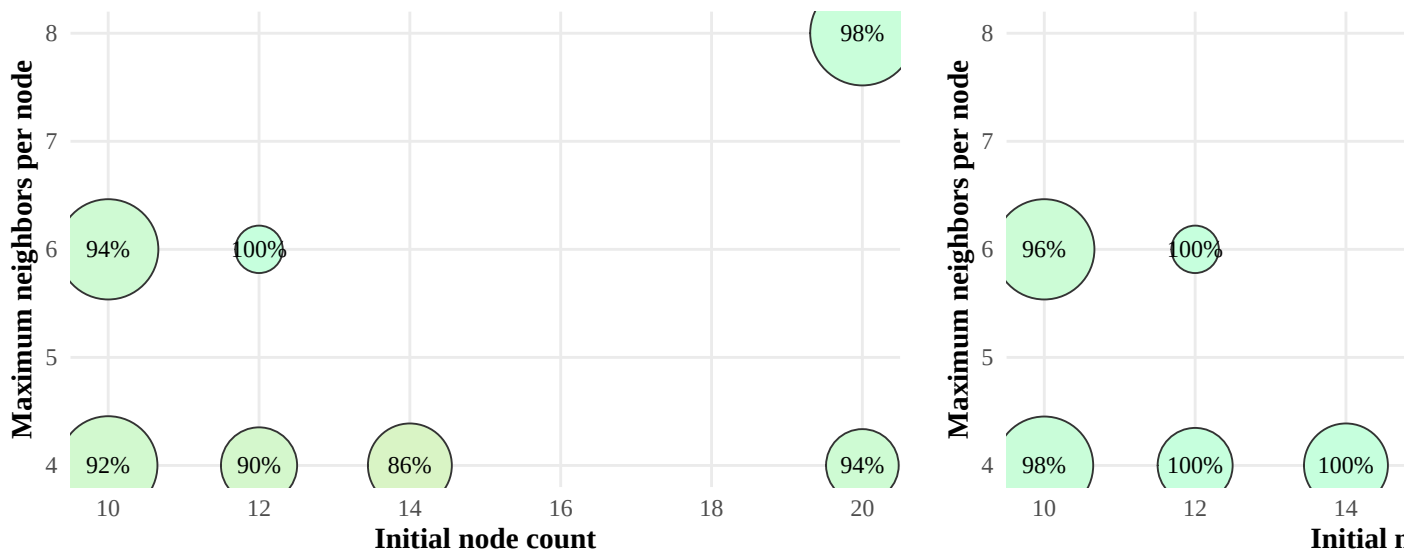
3.5 Combined Input-Variable Comparison

Final Success Rates Across Tested Network Sizes, Neighbor Limits, Node Addition

Final success includes successful replacement-based recovery; bubble size represents the number of re

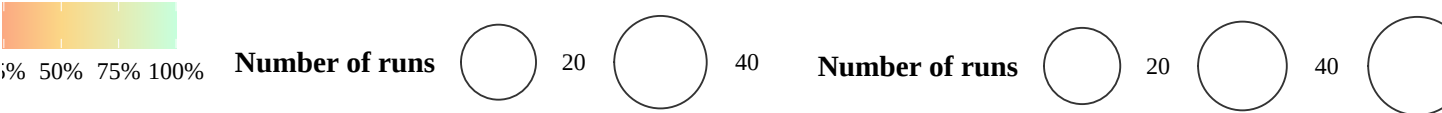
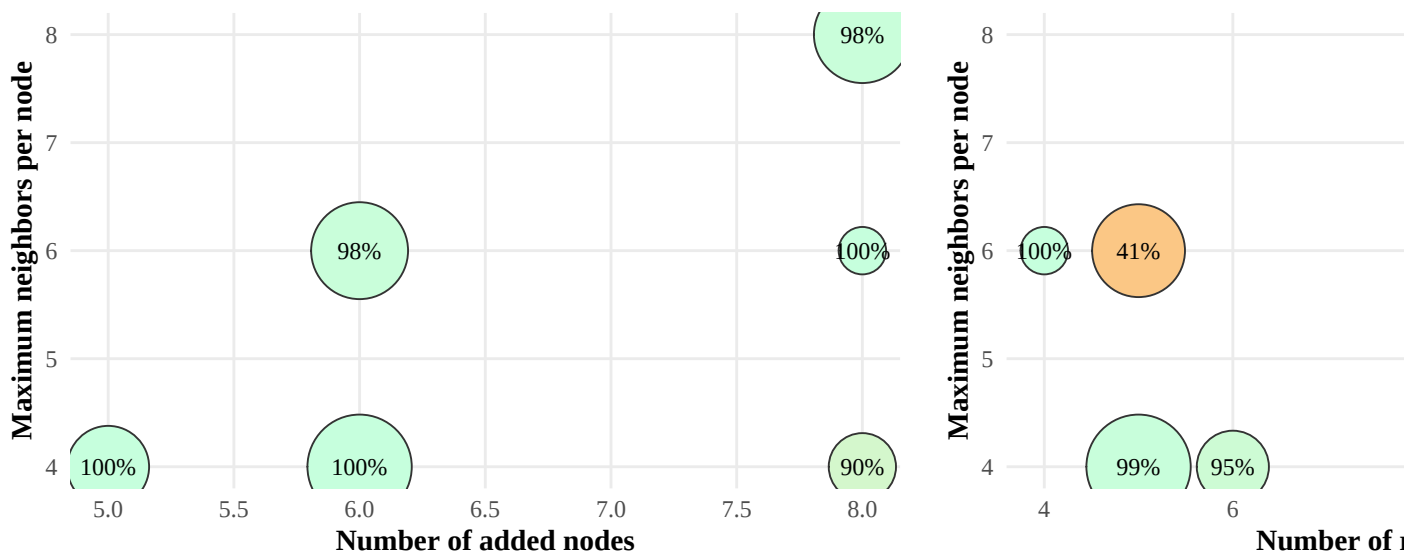
Final Success Rate by Network Size and Neighbor Limit

l and label show final success rate including successful R1 recovery; bubble size represents the number of runs

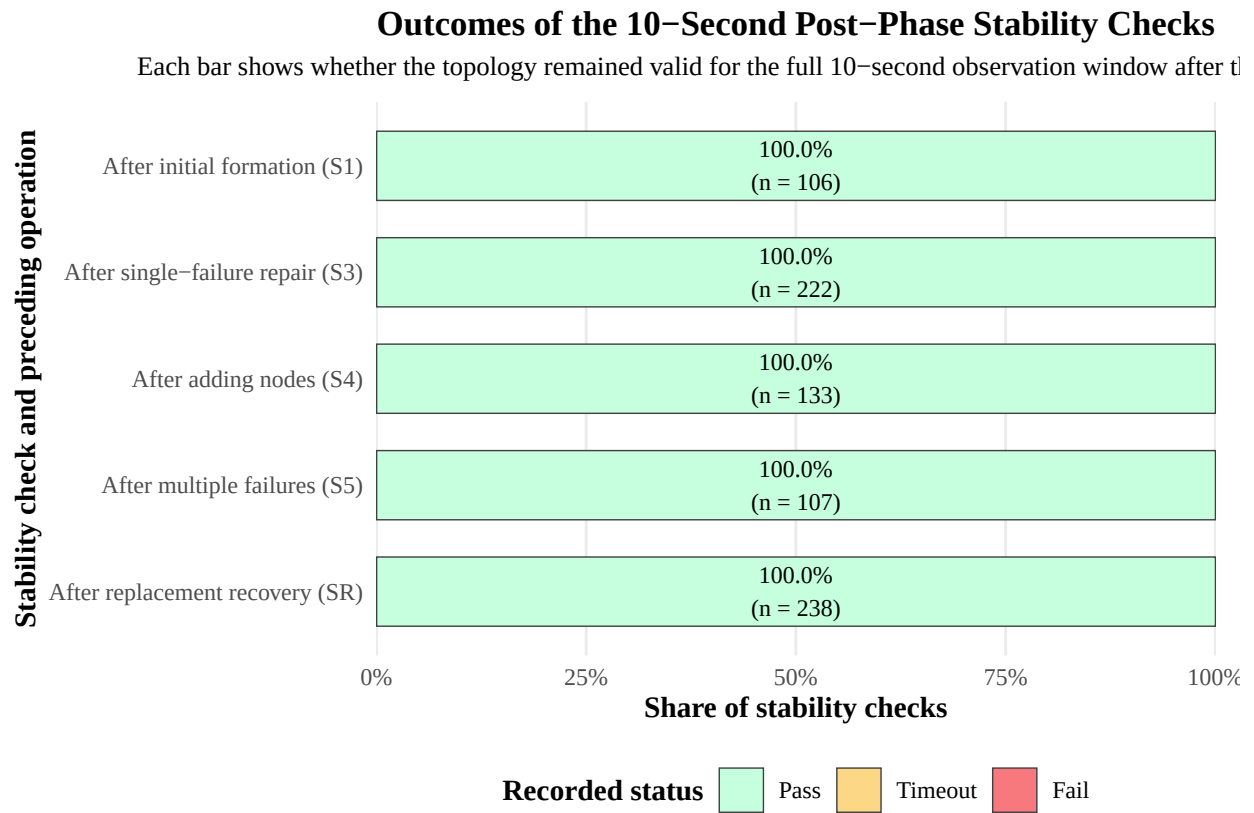


Final Success Rate by Added Nodes and Neighbor Limit

l and label show final success rate including successful R4 recovery; bubble size represents the number of runs



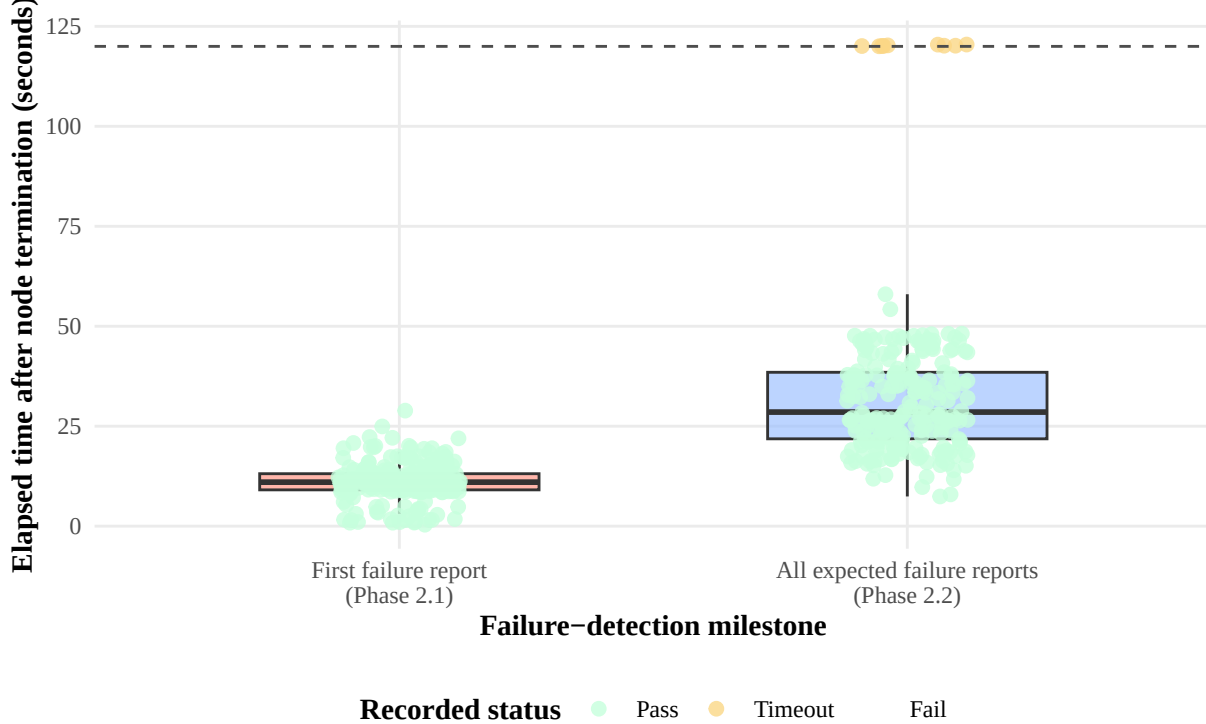
4 Stability Checks



5 Failure Detection Time

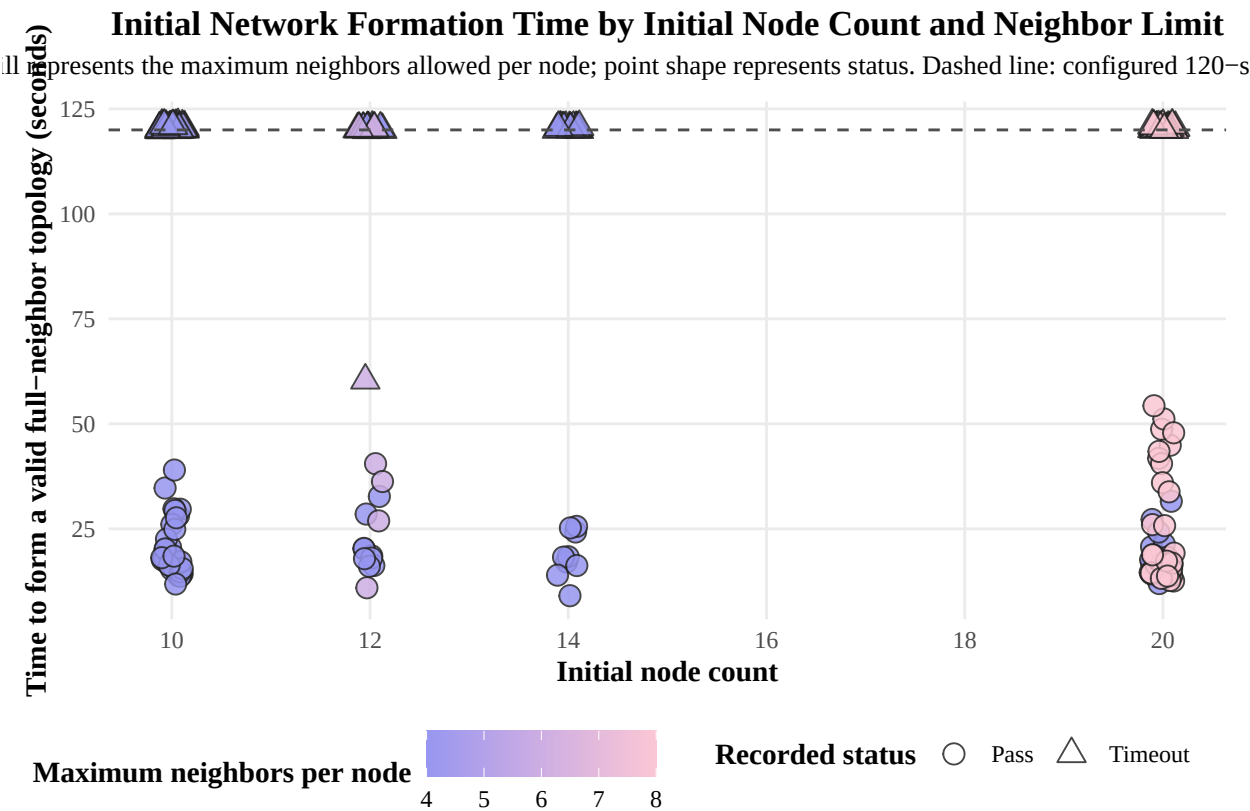
Failure–Detection Time: First Report Compared with Complete Expected Report

ends at the first failure report; Phase 2.2 ends when all expected neighbors have reported. Dashed line: configured 120

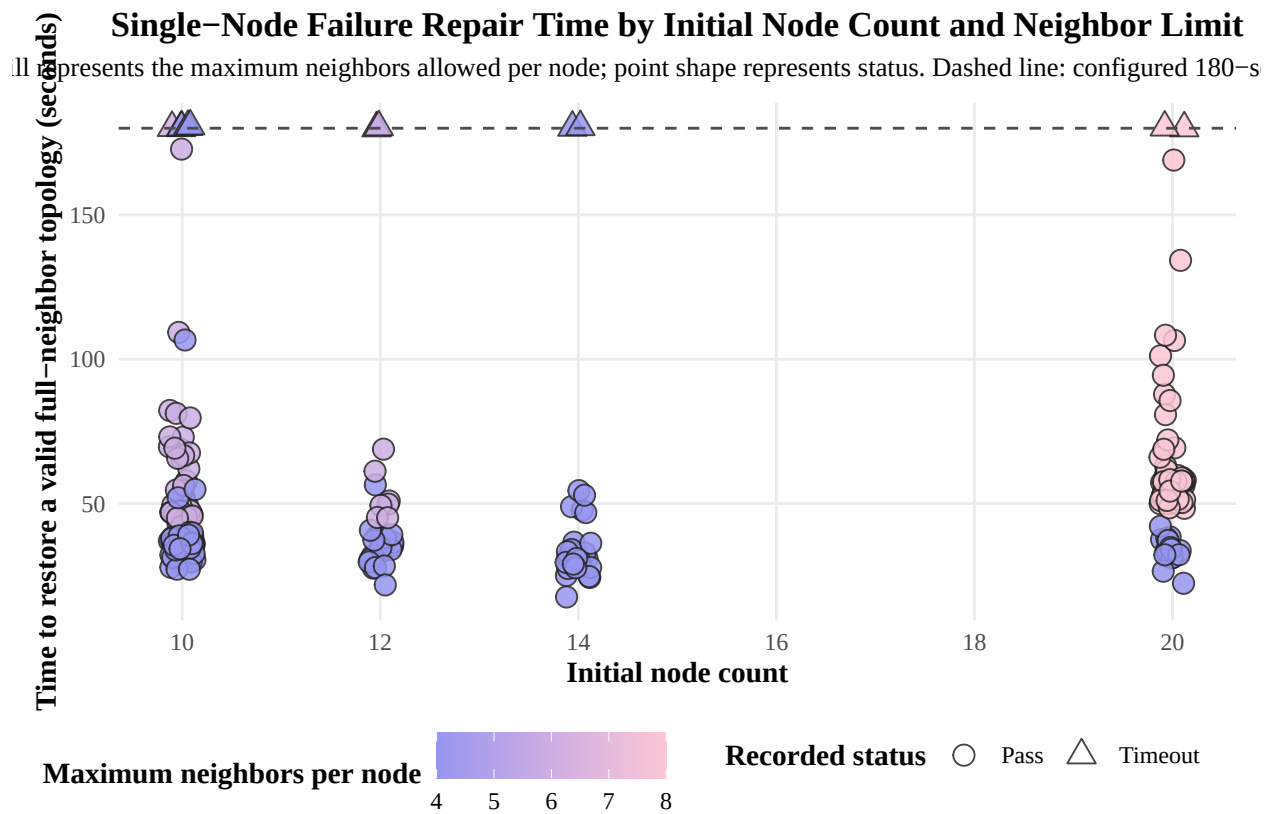


6 Main-Phase Completion Time and Input Effects

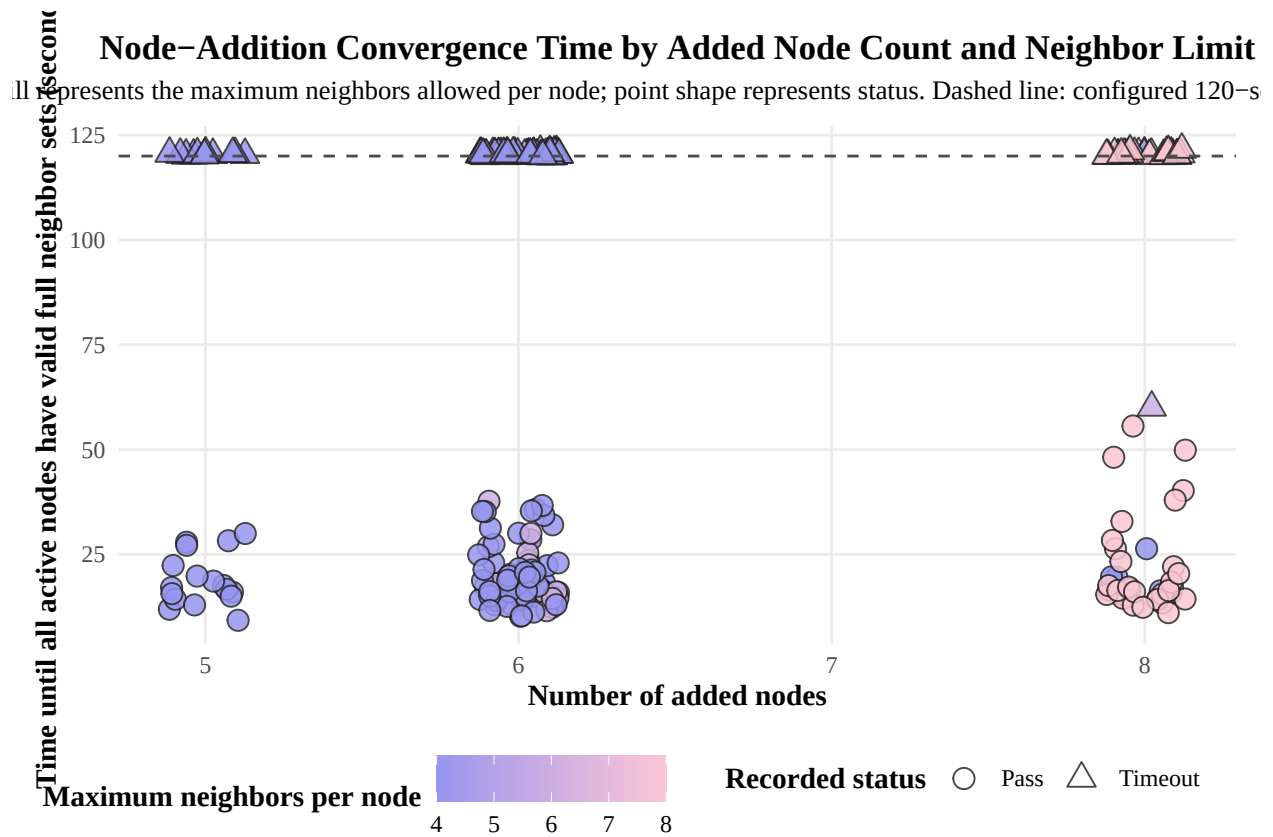
6.1 Phase 1: Initial Network Formation



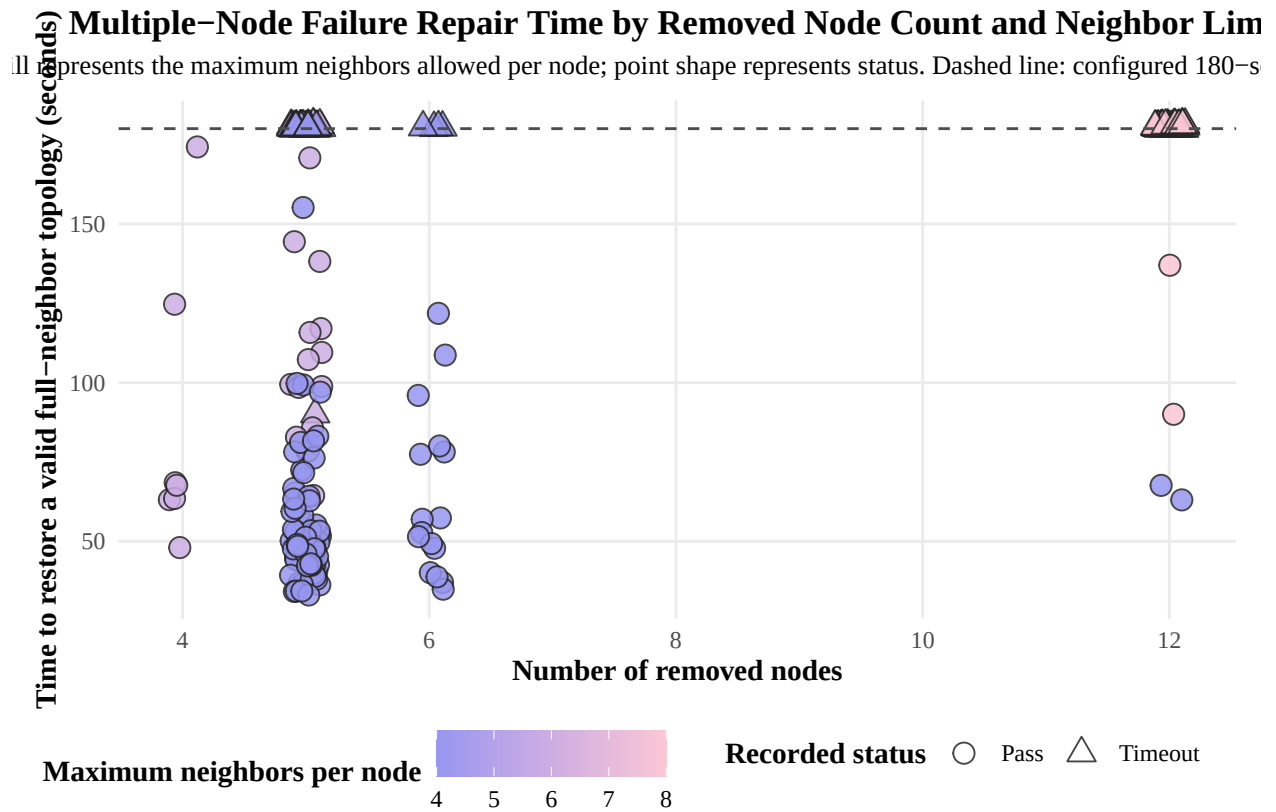
6.2 Phase 3: Repair After One Node Failure



6.3 Phase 4: Adding Nodes



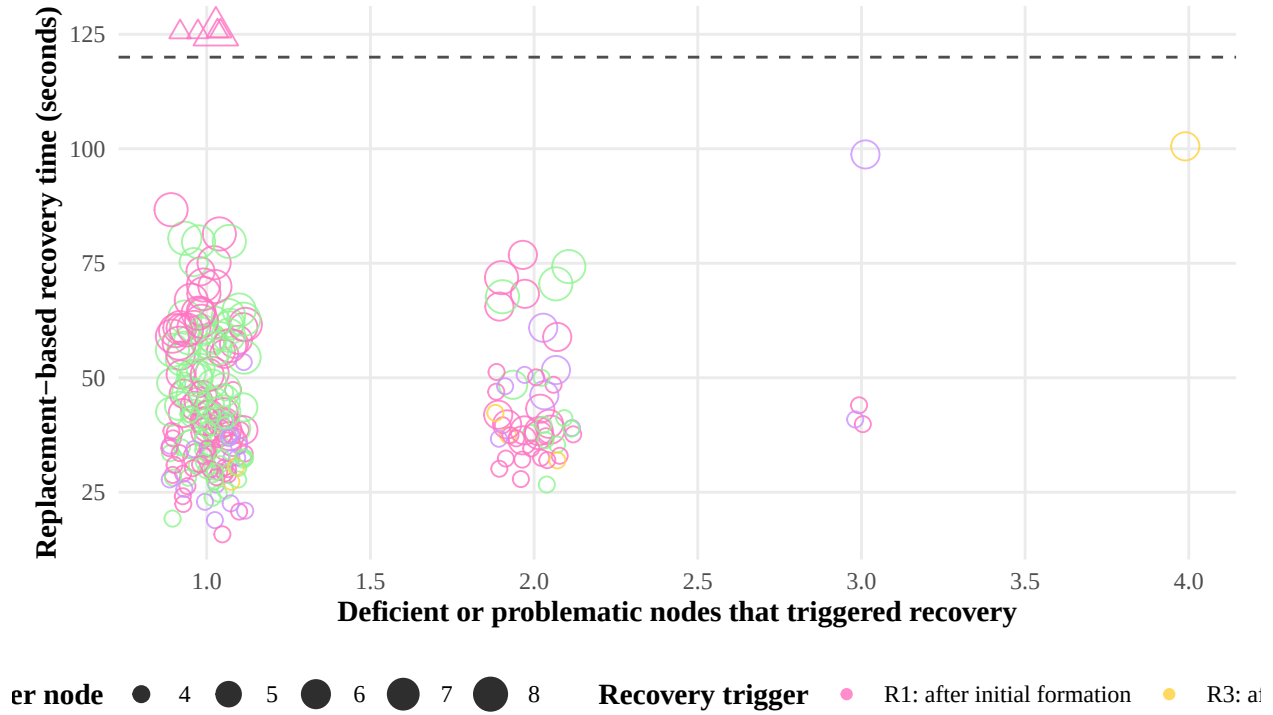
6.4 Phase 5: Repair After Multiple Node Failures



7 Replacement-Based Recovery Completion Time

Replacement-Based Recovery Time by Triggering Operation and Initial Problem C

R4, and R5 restart deficient/problematic nodes and wait for a valid topology. Dashed line: configured 120-second re

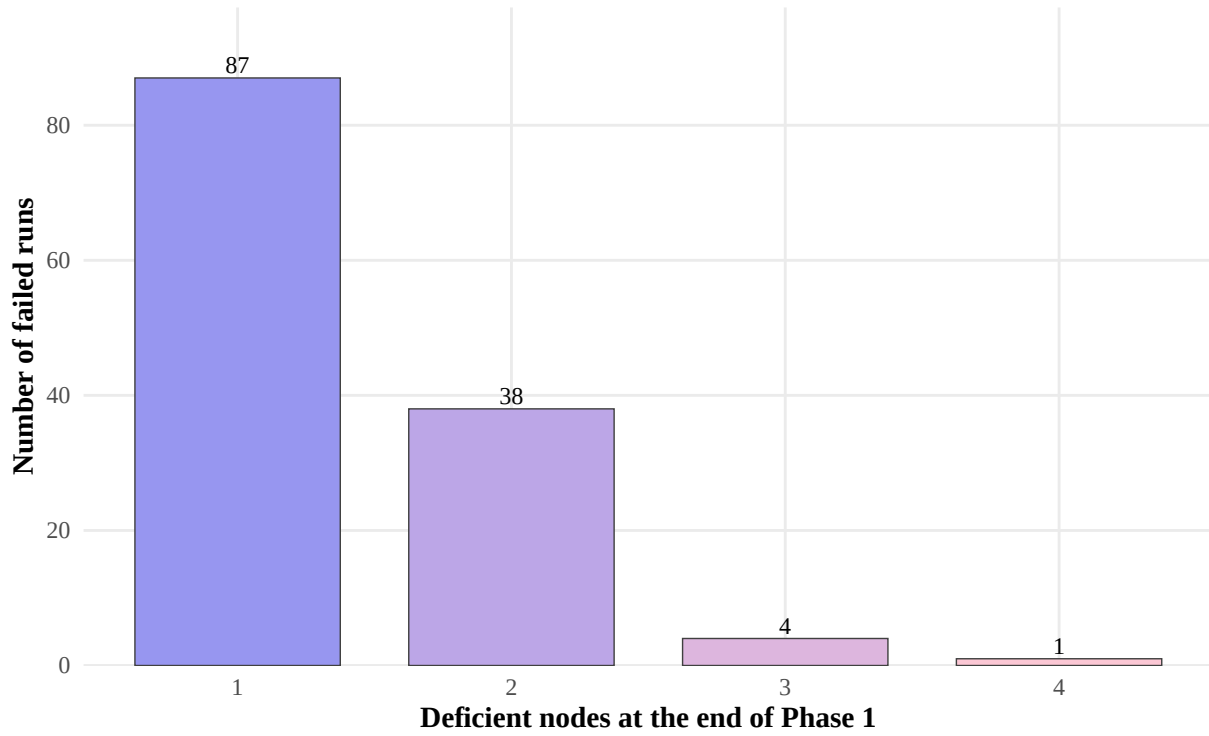


8 Problematic Nodes in Failed Main Phases

8.1 Phase 1: Initial Network Formation

Deficient Nodes Present When Initial Network Formation Failed

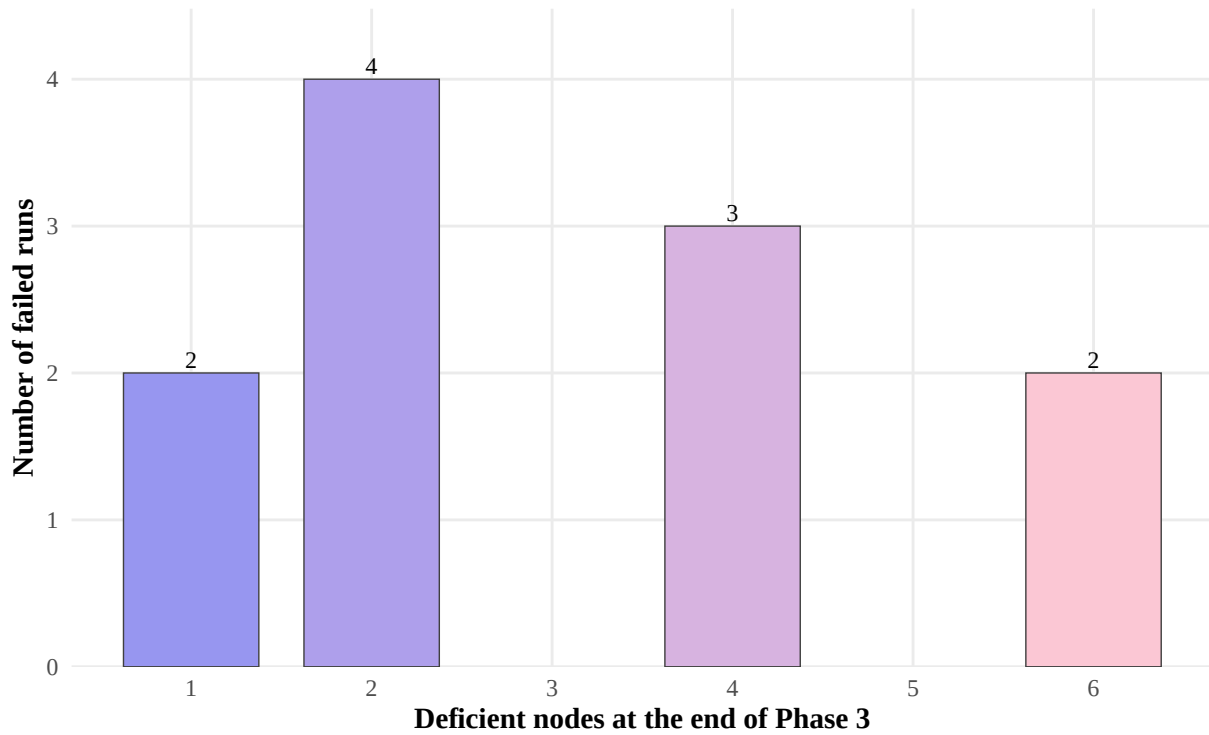
Each bar counts failed Phase 1 runs with the displayed number of deficient nodes.



8.2 Phase 3: Repair After One Node Failure

Deficient Nodes Remaining When Single-Node Failure Repair Failed

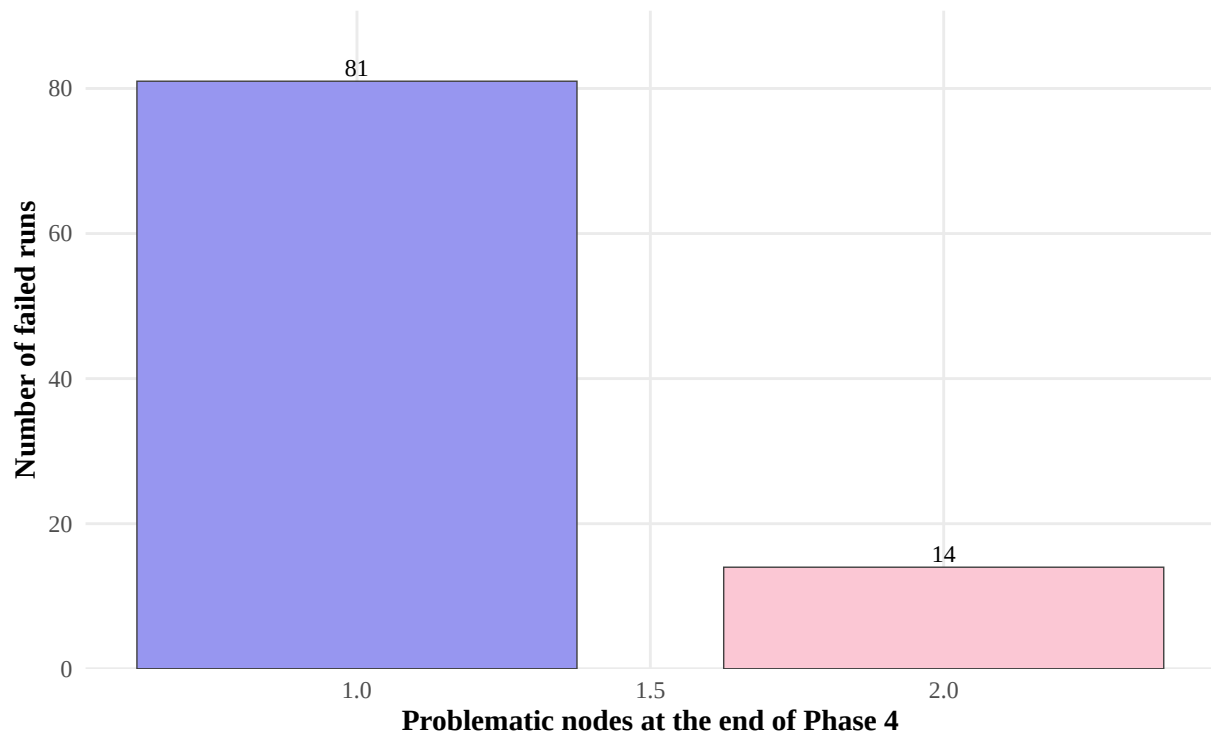
Each bar counts failed Phase 3 runs with the displayed number of deficient nodes.



8.3 Phase 4: Adding Nodes

Problematic Nodes Remaining When Node Addition Failed

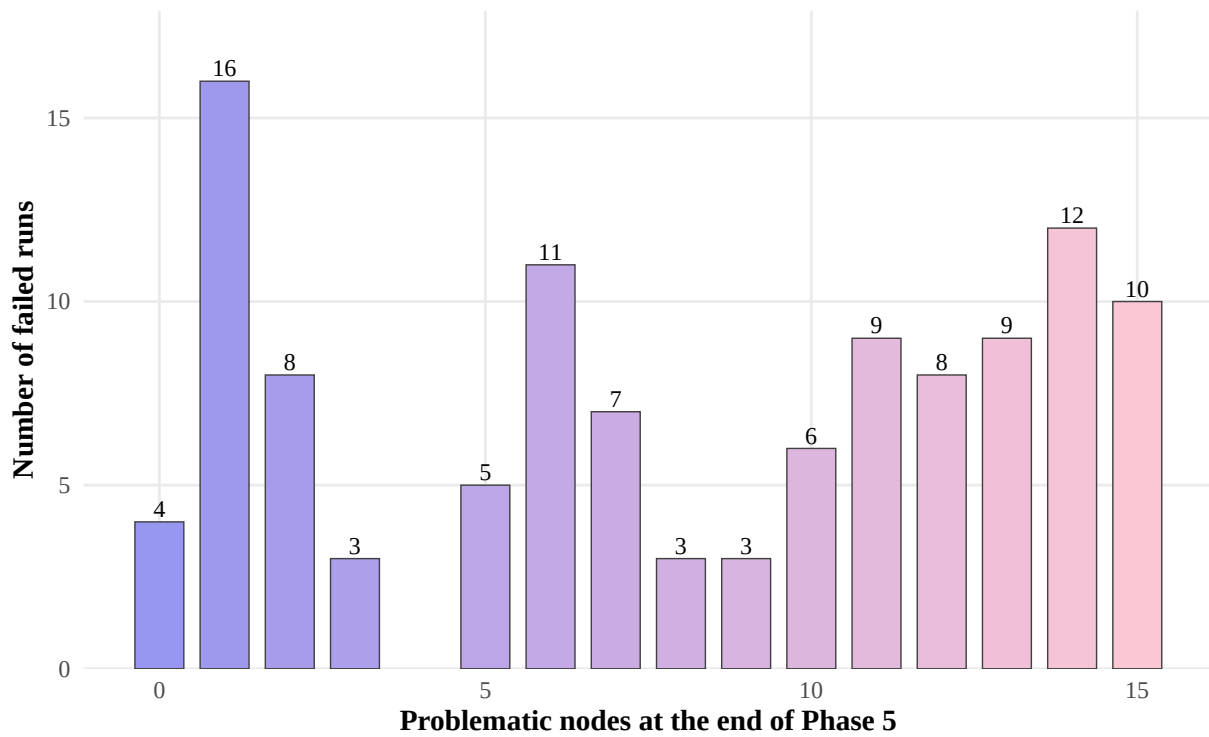
Each bar counts failed Phase 4 runs with the displayed number of problematic nodes.



8.4 Phase 5: Repair After Multiple Node Failures

Problematic Nodes Remaining When Multiple-Node Failure Repair Failed

Each bar counts failed Phase 5 runs with the displayed number of problematic nodes.



9 Replacement-Based Recovery Success Rates

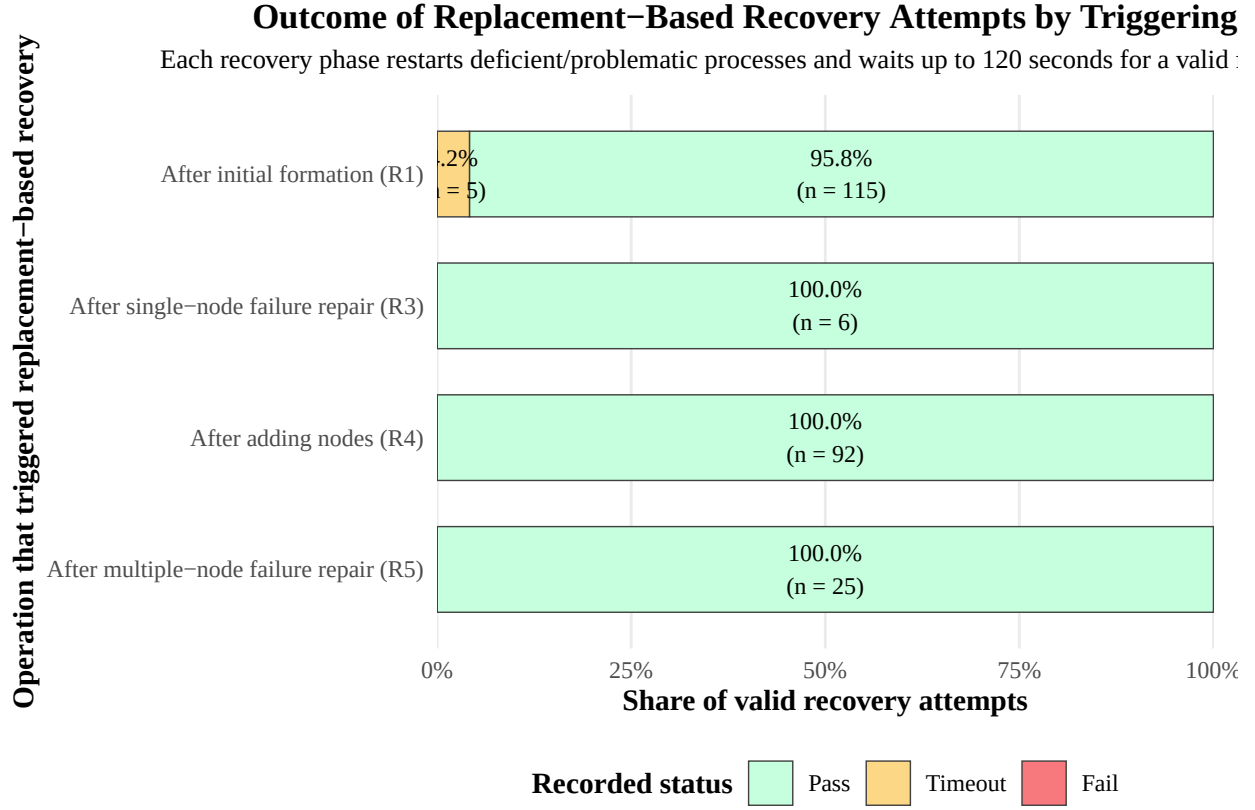


Table 4: Valid replacement-based recovery attempts after timeout filtering.

phase	attempts	passes	timeouts	failures	success_rate
R1	120	115	5	0	0.958
R3	6	6	0	0	1.000
R4	92	92	0	0	1.000
R5	25	25	0	0	1.000